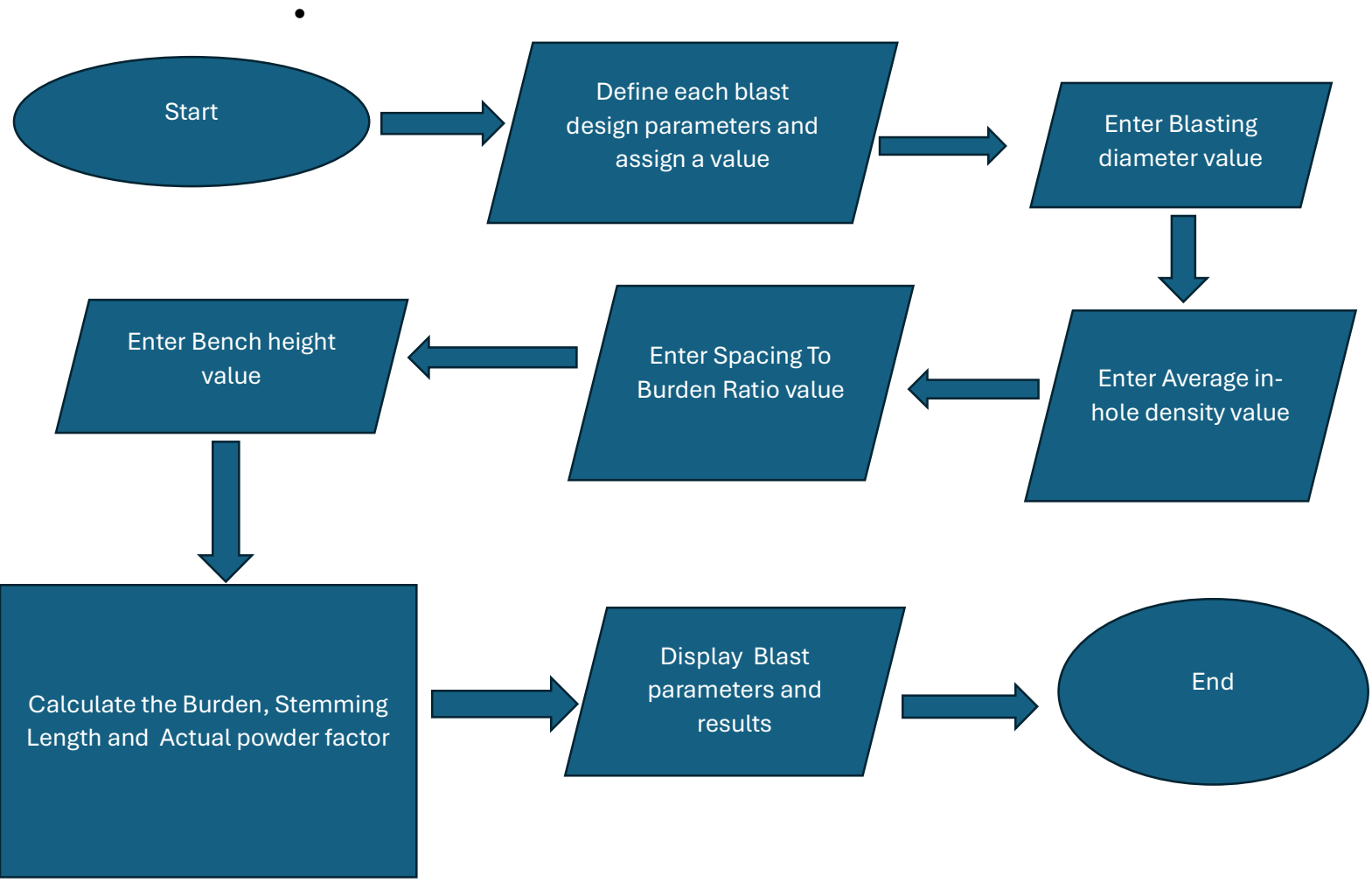
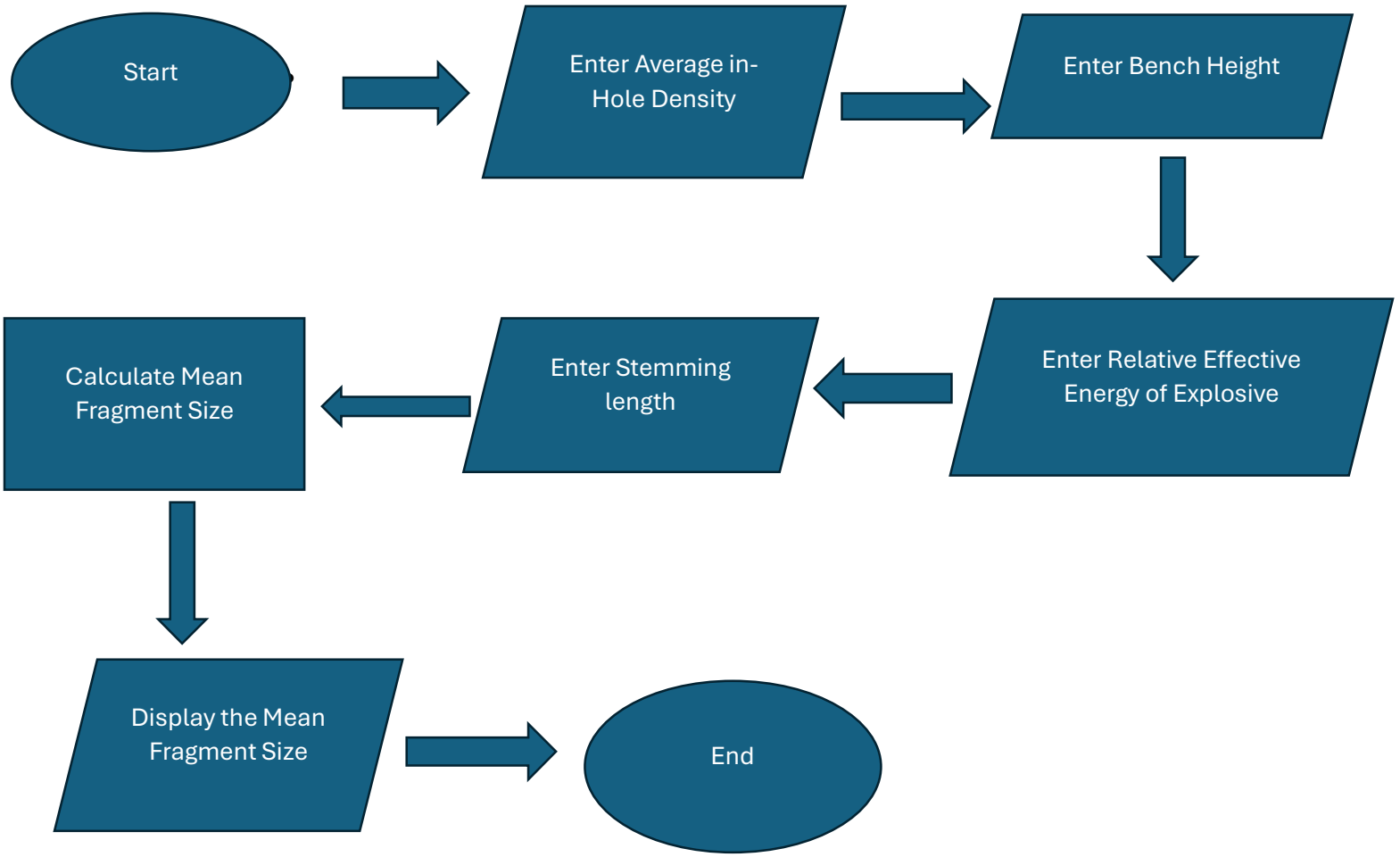
Algorithm for BlastSmart App

1. Surface Blasting Design Calculator
    - Start the process
    - Define the parameters by assigning each to a variable
    - Write down these following parameters: Blasting Diameter, Average In-Hole Density, Spacing to Burden Ratio and Bench Height as inputs.
    - Write the formulas for the following: Stemming Length, Burden and Actual Powder Factor
    - The user is expected to insert or put their own value which will then calculate the required explosive quantity and blast parameters
    - End the process
2. Fragmentation calculations
    - Start the process
    - Define the parameters by assigning each to a variable
    - These parameters (Average In-Hole Density, Bench Height, Relative Effective Energy of Explosive, Stemming Length) should be inserted as inputs
    - Write down the formula to find the mean fragment size
    - The user will put their own value then it will be calculated
    - End the process


Start
Define each blast design parameters and assign a value
Enter Blasting diameter value
Enter Average in-hole density value
Enter Spacing To Burden Ratio value
Enter Bench height value
Calculate the Burden, Stemming Length and Actual powder factor
Display Blast parameters and results
End


Start
Enter Average in-Hole Density
Enter Bench Height
Enter Relative Effective Energy of Explosive
Enter Stemming length
Calculate Mean Fragment Size
Display the Mean Fragment Size
End